27. If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ where $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ then the given pair of linear equation has _____ solution(s)

(1) no solution (2) two solutions (3) unique (4) infinite

28. If $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ where $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ then the given pair of linear equation has _____ solution(s)

(1) no solution (2) two solutions (3) infinite (4) unique

29. GCD of any two prime numbers is _____

(1) -1 (2) 0 (3) 1 (4) 2

30. The GCD of $x^4 - y^4$ and $x^2 - y^2$ is

(1) $x^4 - y^4$ (2) $x^2 - y^2$ (3) $(x + y)^2$ (4) $(x + y)^4$

Points to Remember

- An algebraic expression of the form $p(x) = a_nx^n + a_{n-1}x^{n-1} + ... + a_2x^2 + a_1x + a_0$ is called Polynomial in one variable x of degree 'n' where $a_0, a_1, a_2, ...a_n$ are constants ($a_n \neq 0$) and n is a whole number.
- Let $p(x)$ be a polynominal. If $p(a) = 0$ then we say that 'a' is a zero of the polynomial $p(x)$
- If $x = a$ statisfies the polynominal $p(x) = 0$ then $x = a$ is called a root of the polynominal equation $p(x) = 0$.
- **Remainder Theorem:** If a polynomial $p(x)$ of degree greater than or equal to one is divided by a linear polynomial $(x–a)$, then the remainder is $p(a)$, where a is any real number.
- **Factor Theorem**
  If $p(x)$ is divided by $(x - a)$ and the remainder $p(a) = 0$, then $(x - a)$ is a factor of the polynomial $p(x)$
- Solution of an equation is the set of all values that when substituted for unknowns make an equation true.
- An equation with two variable each with exponent as 1 and not multiplied with each other is called a linear equation with two variables.
- Linear equation in two variables has infinite number of solutions.
- The graph of a linear equation in two variables is a straight line.
- Simultaneous linear equations consists of two or more linear equations with the same variables.

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